

SUJITH MAKAM

AI Developer | Software Engineer

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Bengaluru, India

Summary

AI/Software Engineer with 1+ year building production LLM and agentic-AI systems on Azure and AWS. Delivers RAG pipelines, multi-agent orchestration, and Infrastructure-as-Code contact-center platforms; author of 4 IEEE-published papers in ML and network security.

Education

Indian Institute of Information Technology (IIIT), Dharwad

DEC 2021 – JULY 2025

B.Tech in Computer Science and Engineering

CGPA: 8.41

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, C, SQL

AI & Machine Learning: Generative AI, Agentic AI, LLMs, RAG, Multi-Agent Systems, LLM Evaluation, Prompt Engineering, MCP, A2A Protocol, Google ADK, CrewAI, Azure AI Foundry, MLOps

Backend & APIs: Node.js, Express.js, REST APIs, WebSockets, JSON-RPC, Microservices, Serverless

Cloud & DevOps: AWS (Lambda, Bedrock, S3, DynamoDB, EventBridge, IAM, KMS, CDK, Amazon Connect), Azure (Functions, Cosmos DB), GCP (Cloud Run, Vertex AI), Docker, CI/CD, IaC

Databases: MongoDB, MySQL, PostgreSQL, DynamoDB, Cosmos DB, Vector Databases

Observability & Tools: OpenTelemetry, Langfuse, AgentOps, CloudWatch, React.js, Git, Pytest, Cognigy, Genesys, Linux

Experience

CGI

Jul. 2025 – Present

Software Engineer

- Designed and shipped enterprise **agentic-AI** solutions across Azure AI Foundry, Cognigy, Genesys and AWS, automating customer-support workflows and improving operational efficiency.
- Built production LLM applications (call summarization, automated QA evaluation) with Azure OpenAI and **RAG** over enterprise knowledge bases.
- Developed multi-agent workflows for autonomous task execution, reducing manual intervention and improving SLA compliance.
- Provisioned complete **Amazon Connect** environments as **Infrastructure-as-Code** with **AWS CDK (TypeScript)**, enabling automated multi-region contact-center deployments.

Indian Institute of Science, Bangalore

May. 2024 – July. 2024

Research Intern

- Profiled GPT-2 with Intel VTune to locate attention and feed-forward bottlenecks, achieving a **4x speedup** in attention-layer execution.
- Optimized those layers via custom tiled matrix multiplication, cutting memory bandwidth usage by **34.8%** and instruction-cache misses by **5.2%**.

Projects

LegacyLift | Multi-Agent, MCP, A2A, Next.js



- Built a multi-agent system making legacy **COBOL** logic callable by AI agents without rewriting the mainframe; **top 13 of 200+ entries at CGI Envision 2026**.
- Implemented 5 collaborating agents over an **A2A** message bus with a shared context graph, parsing COBOL source to expose routines and auto-generate **MCP** tools.
- Streamed live execution state to a Next.js UI over a WebSocket relay behind a provider abstraction to avoid vendor SDK coupling.

AI Governance Platform | Azure AI Foundry, Google ADK, CrewAI



- Built an AI governance platform for multi-agent workflows using Azure AI Foundry, Google ADK, CrewAI and Agent Gateway for secure LLM-agent orchestration.
- Integrated Langfuse and AgentOps for agent observability, prompt tracing, KPI monitoring and latency analysis.
- Implemented Responsible-AI controls: prompt-injection prevention, sensitive-data masking, API isolation and policy enforcement.

Certifications

CGI: Inside Agentic AI (Foundations; Core Architecture), AI Guardrails & Governance, Elements of an AI Architect, Reusable AI Architecture Patterns, Cloud Computing & MLOps, Microsoft Foundry for Enterprise AI

Udemy: Full Stack Web Development, Machine Learning A-Z

Research Publications

- FAPL-DM-BC: Adaptive Privacy-Aware, Blockchain-Backed FL Framework with Dual-Model XAI for the IoVs – **IEEE COMSNETS 2026**
- Enhancing IoT Network Security through Adaptive Curriculum Learning and XAI – **IEEE FNWF 2025**
- FedGuard-IIoT: A Scalable Federated Learning Architecture for Network-IDS in the Industrial IoT – **IEEE FNWF 2025**
- Predicting Stellar Metallicity: A Comparative Analysis of Regression Models for Solar Twin Stars – **IEEE SPACE 2025**

Achievements

- **CGI Envision 2026:** Top 13 of 200+ submissions (LegacyLift).
- **Andhra Pradesh State Board Topper:** 990/1000.